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Switched NMPC for epidemiological and social-economic control objectives in SIR-type systems

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ABSTRACT

Several optimal control strategies have recently been developed to minimize the infected peak prevalence or the epidemic final size. Although these two indexes are critical to assess any control policy tending to mitigate an epidemic by means of non-pharmaceutical measures, they are usually considered separately and, in general, no consensus has been reached about how to simultaneously handle them in a simple and realistic way (i.e., accounting for the limitations in the control actions, avoiding new cycles of infections or reboundings, considering side effects, etc.) Here, based on a theoretical dynamical analysis of SIR-type models, a realistic nonlinear model predictive control strategy is proposed. Apart from minimizing the epidemic final size and keeping the infected peak prevalence under an established value, the controller accounts for feedback uncertainty and different actuator constraints, such as a limited number of social distancing policies, which may remain active for a minimal and a maximal time interval. Several simulations considering different SIR-type models illustrate the benefits of the proposal.

1. Introduction

Lessons learned from recent global epidemics taught us the important balance between healthcare system security and economic sustainability. Indeed, the COVID-19 pandemic had a profound impact on nearly every aspect of human life, from health and economy to social and political interactions, making it one of the most significant events in recent history (Bashir, Ma, & Shahzad, 2020; Brodeur, Gray, Islam, & Bhuiyan, 2021; Lamers & Haagmans, 2022). Consequently, the scientific community has supported the policy decision-making process (Alamo, Millán, Reina, Preciado, & Giordano, 2021; Alamo, Reina, Gata, Preciado, & Giordano, 2021; Hernandez-Vargas et al., 2020). One of the earliest ways to deal with the pandemic was the implementation of the so-called Non-Pharmaceutical Interventions (NPIs), such as: banning of gatherings, university and school closures, stay-at-home measures, closing non-essential businesses, mask-wearing requirements, among others Flaxman et al. (2020). These measures are considered as different levels of Social Distancing (SD) and allowed policy-makers to lessen disease transmission. Nevertheless, until now there is no clear consensus on the optimal approach to applying these intervention measures since this issue is usually treated as a trade-off Optimal Control Problem (OCP) in a variety of proposals reported in the literature.

This trade-off comes from posing the OCP formulation in such a way that results in competition between healthcare and socio-economic interests due to the application of NPIs. Evidence shows that considering straightforward control objectives, i.e., minimizing the number of infected/decease individuals at every time, while minimizing the cost of SD measures, largely produces sub-optimal solutions (Abbasi, 2020; Punzo, 2022; Sereno, Anderson, Ferramosca, Hernandez-Vargas, & González, 2022). Properly posing an optimal control problem requires knowledge of the system dynamics to know what can (and more importantly, what cannot) be done with the system under control. Asking for an impossible task, and ignoring the limits that the system imposes, usually leads to poor performance. In the particular case of SARS-CoV-2, thinking that it is possible to fully control and/or manipulate a global epidemic by external intervention is, at best, a naive illusion and, at worst, a dangerous belief, since it motivates to take severe measures that produce the exact opposite effect to the one sought (Abbasi, 2020; Hale et al., 2021; Rypdal, Bianchi, & Rypdal, 2020).

To measure the impact of epidemic outbreaks two metrics are commonly used: the Infected Peak Prevalence (IPP, which corresponds to the maximal fraction of infected individuals over the total pandemic

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time) and the Epidemic Final Size, (EFS, total fraction of infected individuals at the end of the pandemic outbreak). Both, the minimization of the IPP (Morris, Rossine, Plotkin, & Levin, 2021; Sadeghi, Greene, & Sontag, 2021) and the EFS (Bliman & Duprez, 2021) were proposed as control objectives in the context of single-interval policies; i.e., simple control strategies consisting in a fixed control action that reduces the transmission rate for a finite period of time. One step ahead are the optimal control strategies, which consider a more complex sequence of control actions, together with explicit constraints on both, manipulated and controlled variables.

Extensive literature on optimal control strategies for the design of SD measures could be found. Among many others, the Model Predictive Control (MPC) approach is one of the most applied strategies. As we can see in Morato et al. (2020), an MPC based on SIR-type models was proposed. The control objective consists in minimizing both, the number of infected individuals and the time of isolation, while some constraints account for the peak of the infected fraction, and the minimal time of application of the control actions. In Köhler et al. (2021), a robust nonlinear MPC based on the SIDARTHE model, introduced in Giordano et al. (2020), is proposed. The MPC manipulates also the transmission rate, given in this case by several parameters. The objectives are to minimize both, the number of fatalities and the time of isolation, compared to a baseline policy. The authors consider that the control policy updates every 7 day, and is allowed to take continuous values between the range of $[0, 1]$, where 0 represents no SD, and 1 represents the hardest lockdown measure.

In Carli, Cavone, Epicoco, Scarabaggio, and Dotoli (2020), a Nonlinear Model Predictive Control (NMPC) strategy for a multi-region SIR-type model was proposed. The formulation minimizes the use of NPIs, while the healthcare capacity is considered a soft constraint of the objective function. The authors consider 3 different levels of SD for each region and an on–off decision variable for the travel mobility between them. The control actions are updated every week. In Péni, Csutak, Szederkényi, and Röst (2020) a NMPC with logic constraints is proposed to manage the COVID-19 epidemic with the possibility to handle time-dependent constraints, logic relations between model variables, and predefined discrete levels of interventions. The cost function is set to minimize the effect of control measures (mitigation scenario) and also the hospitalized and deceased population (suppression scenario). Both, the healthcare system capacity and the control action have upper-bound constraints. Similarly to Carli et al. (2020) the authors consider the effect of control input quantization, thus allowing to consider fixed realistic levels of SD measures. Later, the authors extended the strategy to allow computationally efficient solutions (Péni & Szederkényi, 2021). In both proposals, (Péni et al., 2020; Péni & Szederkényi, 2021), the control input is updated every week. In Pataro et al. (2021) an NMPC is proposed to control the COVID-19 epidemic in Salvador de Bahia, Brazil. The authors consider degrees of population compliance with the SD measures and a trade-off objective function accounting for the minimization of infected cases and SD measures. Similarly to the works discussed previously, the control signal is updated every week.

Other recent control-based strategies can be seen in Berger (2022), Dewasme and Vande Wouwer (2022), Dias, Queiroz, and Araujo (2022) and Kasis, Timotheou, Monshizadeh, and Polycarpou (2022). None of the reported studies consider the minimization of the EFS under finite-time interruption of the measures, in a way that avoids second infection waves, as stated in Sereno et al. (2022). Although some of the work discussed here consider the quantization/discretization of the control action according to discrete levels of NPIs (Carli et al., 2020; Pataro et al., 2021; Péni et al., 2020), none of them consider the minimum and maximum time of application for each of the possible SD levels. Updating social distancing measures every week, as many papers propose, may actually be impractical for many policymakers, and may even cause social fatigue. To mimic real-life scenarios is necessary to extend the control policy update to allow minimal and maximal time of application for each possible SD measure, i.e., for soft SD measures longer periods of time could be allowed, while for hard SD measures, short periods of time should be imposed.

1.1. Work contributions

The objective of this article is to propose a realistic control strategy to close the loop of SIR-type epidemiological control systems, by exploiting the theoretical results presented in Sereno et al. (2022), and taking into account specific limitations for the control action. A Switched Nonlinear Model Predictive Control (sNMPC) strategy is proposed, which minimizes the epidemic final size, maintains the infected peak prevalence under a value imposed by the healthcare system capacity, and minimizes – as long as possible – the NPI severity. A discretization of the possible NPI's is made, and constraints on the maximal and minimal time of application of each intervention are explicitly imposed. The closed-loop robustness is assessed by simulating moderate-to-severe input/output disturbances, and model uncertainties for different SIR-type systems (including the so-called SIDARTHE, Giordano et al. (2020)). The obtained results are essentially different from most control-based approaches found in the literature (Köhler et al., 2021; Morato et al., 2020; Péni & Szederkényi, 2021).

2. SIR-type modeling

Consider the typical SIR epidemic model[1] Kermack and McKendrick (1927), which describes the fractions of susceptible $S(t)$ and infectious $I(t)$ individuals in a population at time t. Infections occur proportional to $S(t)I(t)$ at a transmission rate β, and infectious individuals recover or die at a rate γ (the compartment for removed, R, is not considered here). Non-pharmaceutical interventions are assumed to reduce the effective transmission rate, $\beta(t)$, below its value in the absence of intervention, which is considered fixed. By rescaling the time by $\tau := t\gamma$, the SIR model can be written in non-dimensional form as (Bertozzi et al., 2020; Sereno et al., 2022; Sontag, 2011):

$$\begin{aligned} \dot{S}(\tau) &= -\mathcal{R}(\tau)S(\tau)I(\tau), \\ \dot{I}(\tau) &= \mathcal{R}(\tau)S(\tau)I(\tau) - I(\tau), \end{aligned} \tag{1}$$

where $\mathcal{R}(\cdot) := \beta(\cdot)/\gamma$ denotes the time-varying reproduction number, the epidemic outbreak time is assumed to be at $\tau = 0$, and $(S(0), I(0)) := (1-\epsilon, \epsilon)$, with $0 < \epsilon \ll 1$; *i.e.*, the fraction of susceptible individuals is smaller than, but close to 1, while the fraction of infectious is close to zero. Parameter $\mathcal{R}(\cdot)$ is directly affected (reduced by some amount) by NPIs, so it is assumed that it has a maximum $\overline{\mathcal{R}} > \frac{1}{1-\epsilon} > 1$, which is the reproduction number in absence of interventions (and it is greater than one since otherwise there is not an epidemic outbreak), and a minimum $\underline{\mathcal{R}} > 0$, which denotes the maximal effectiveness of interventions (hard quarantines, for instance) and is assumed to be greater than zero (perfect quarantines are not possible). Furthermore, we define two times $0 < \tau_i < \tau_f < \infty$ which denotes the initial and final intervention times and τ_f is assumed to be finite to account for realistic social intervention planning.

According to real government policies, the NPI severity is not a continuous variable, but a discrete one, assuming just a few possible values from the hardest one (lockdown, quarantines), to more relaxed ones (closing public spaces, public transportation, schools, face mask-wearing, etc.), to the complete absence of measures. To represent this characteristic of the control problem, we express the reproduction number as

$$\mathcal{R}^{\sigma}(\tau) = \overline{\mathcal{R}} - \sigma(\tau)\Delta\mathcal{R}, \tag{2}$$

with

$$\Delta\mathcal{R} = \frac{\overline{\mathcal{R}} - \underline{\mathcal{R}}}{n},$$

[1] Note that any extension of SIR basic models – such as the SIRD, the SIRASD (Morato et al., 2020) and the SIDARTHE (Giordano et al., 2020) – shows the same critical behavior (as detailed in Sadeghi et al. (2021)), given that they only made a subdivision of each of the infected compartment. This is demonstrated in the Simulation Section 4.

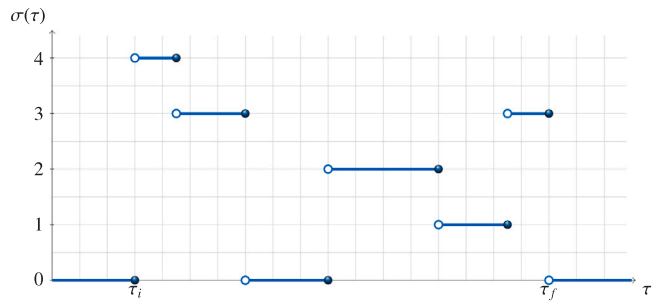

Fig. 1. Schematic plot of the intervention pattern $\sigma(\tau)$, when the number of possible intervention levels is $n = 4$.

Table 1
Levels of NPI severities, for $n = 4$.

$\sigma(\tau)$	NPI	$\mathcal{R}^\sigma(\tau)$
0	no NPI	$\overline{\mathcal{R}}$
1	addition of mask-wearing	$\overline{\mathcal{R}} - \Delta\mathcal{R}$
2	addition of university/school closure	$\overline{\mathcal{R}} - 2\Delta\mathcal{R}$
3	addition of stay-at-home measures	$\overline{\mathcal{R}} - 3\Delta\mathcal{R}$
4	hardest lockdown (all available measures)	$\underline{\mathcal{R}}$

where $n \in \mathbb{I}$ is the number of possible NPI severities, and $\sigma(\tau) \in \mathbb{I}_{[0,n]}$ is the control input (piece-wise constant *intervention pattern*) describing the severity of the NPI ($\sigma = 0$ corresponds to no intervention, while $\sigma = n$ corresponds to the hardest lockdown). To account for the initial and final times, τ_i and τ_f, σ is assumed to belong to the following set:

$$\Omega^\sigma := \{\sigma(\cdot) : \mathbb{R}_{[0,\infty)} \to \mathbb{I}_{[0,n]} : \sigma(\tau)=0, \text{ for } \tau\in[0,\tau_i)\cup(\tau_f,\infty)\}.$$

Later on, in the simulation Section 4, constraints on the minimum and maximum time of application of each measurement (waiting-time constraints) will be explicitly considered to account for real limitations.

A schematic plot of $\sigma(\tau)$ is depicted in Fig. 1. Accordingly, $\mathcal{R}^\sigma(\tau)$ is a piece-wise constant function, in such a way that for $\sigma = 0$ no intervention is implemented, and for $\sigma = n$ the hardest lockdown takes place. A particular assignment of the different severity levels of NPIs is shown in Table 1, corresponding to the one made by the Italian Ministry of Health during the SARS-CoV-2 pandemic, in which the regions and autonomous provinces were classified into four areas – red, orange, yellow, and white – according to three risk scenarios (Italian Ministry of Health, 2022).

To make the SIR system explicitly dependent on the control action σ, model (1) is rewritten as

$$\begin{aligned}\dot{S}(\tau) &= -\mathcal{R}^\sigma(\tau)S(\tau)I(\tau),\\ \dot{I}(\tau) &= \mathcal{R}^\sigma(\tau)S(\tau)I(\tau) - I(\tau),\end{aligned} \tag{3}$$

which can be seen, given the nature of the intervention pattern $\sigma(\tau)$, as a special case of a switched system (i.e., a system in which the control action consists in selecting one of n possible subsystems, Anderson, Gonzalez, Ferramosca, and Hernandez-Vargas (2020), Anderson, González, Ferramosca, and Hernandez-Vargas (2021)). Finally, susceptible S and infectious I are constrained to be in the set

$$\mathcal{X} := \{(S, I) \in \mathbb{R}^2 : S \in [0, 1], I \in [0, 1], S + I \le 1\},$$

for all $\tau \ge 0$.

2.1. Equilibrium and system behavior characterization

A control equilibrium is, by definition, an equilibrium of the system (3) associated with some control action $\sigma(\cdot) \in \Omega^\sigma$. However, the intervention pattern is by definition a finite-supported function (it is different from zero in a finite domain given by $[\tau_i, \tau_f]$), so equilibria (i.e., states at which the system can remain indefinitely) can only be considered for $\tau > \tau_f$, when $\sigma(\tau) = 0$ or, the same, for $\mathcal{R}^\sigma(\tau) = \overline{\mathcal{R}}$ (for $\sigma(\cdot) \in \Omega^\sigma$, the system is in transient regime before τ_f). Therefore, the equilibrium of system (3), for $\tau \in (\tau_f, \infty)$ and initial conditions $(S(\tau_f), I(\tau_f)) \in \mathcal{X}$, is obtained by zeroing the differential equations, with $\mathcal{R}^\sigma(\tau) \equiv \overline{\mathcal{R}}$. This gives:

$$\mathcal{X}_s := \{(\bar{S}, \bar{I}) \in \mathcal{X} : \bar{S} \in [0, 1], \bar{I} = 0\}.$$

This set divided into two subsets, $\mathcal{X}_s = \mathcal{X}_s^{st} \cup \mathcal{X}_s^{un}$, where

$$\mathcal{X}_s^{un} := \{(\bar{S}, \bar{I}) \in \mathcal{X} : \bar{S} \in (S^*, 1], \bar{I} = 0\},$$

is an unstable equilibrium set (which contains the healthy equilibrium), and

$$\mathcal{X}_s^{st} := \{(\bar{S}, \bar{I}) \in \mathcal{X} : \bar{S} \in [0, S^*], \bar{I} = 0\},$$

is the unique asymptotically stable (AS) set (see Theorem 1 in Sereno et al. (2022)), with a domain of attraction given by $\mathcal{X}$ and S^* being the *herd immunity*, defined as

$$S^* := 1/\overline{\mathcal{R}}.$$

Any compact set including an AS equilibrium set is AS, so $\mathcal{X}_s$ is AS, for any value of $\overline{\mathcal{R}}$. However, it contains an unstable equilibrium set, $\mathcal{X}_s^{un}$. Figs. 2 shows a Phase Portrait for system (3), for different values of $(S(\tau_f), I(\tau_f)) \in \mathcal{X}$. It also shows the level curves of a Lyapunov function $V(S, I) := S - S^* - S^* \ln(S/S^*) + I$, centered at $(S^*, 0)$.

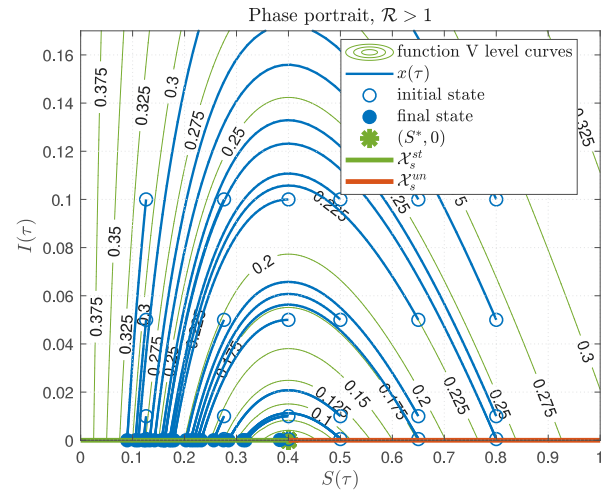

Fig. 2. Phase Portrait for system (3) with $\overline{\mathcal{R}} = 2.9$, for $\tau > \tau_f$ and different values $(S(\tau_f), I(\tau_f)) \in \mathcal{X}$. Set $\mathcal{X}_s^{st}$ is in green (thick line), while $\mathcal{X}_s^{un}$ is in red (thick line). The empty circle denote $(S(\tau_f), I(\tau_f))$, while the filled circles denote $(S(\tau), I(\tau))$ for $\tau \to \infty$. Furthermore, the level curves of a Lyapunov function V are plotted, which constitute invariant trajectories for the system (green thin line). As it can be seen, all the trajectories converge to $\mathcal{X}_s^{st}$, even when no points of $\mathcal{X}_s^{st}$ are attractive, Sereno et al. (2022).

2.2. Epidemic severity indexes

Next, we define the two main epidemic indexes used to assess the severity of the epidemic. According to the outbreak conditions, $(S(0), I(0)) := (1 - \epsilon, \epsilon)$, with $0 < \epsilon \ll 1$, and recalling that $\overline{\mathcal{R}} > \frac{1}{1-\epsilon}$, $I(\tau)$ starts growing up from $\tau = 0$ to τ_i, when $\sigma(\tau) = 0$ ($\mathcal{R}^\sigma = \overline{\mathcal{R}}$), and then will show a behavior depending on the form of $\sigma(\cdot) \in \Omega^\sigma$. On the other side, $S(\tau)$ will be strictly decreasing for $\tau \ge 0$, independently of the form $\mathcal{R}^\sigma(\tau)$. Even more, once $S(\tau)$ goes below the herd immunity S^*, I cannot longer grow up (given that $\mathcal{R}^\sigma(\tau) \le \overline{\mathcal{R}}$ for all $\tau \ge 0$, $\dot{I} = (\mathcal{R}^\sigma(\tau)S(\tau) - 1)I(\tau) \le (\overline{\mathcal{R}}S(\tau) - 1)I(\tau)$ so once $S(\tau) < S^*$, $(\overline{\mathcal{R}}S(\tau) - 1)$ becomes negative and remains so in the future, which means that $\dot{I} < 0$).

In any case, the first epidemic-severity index, i.e., the infected peak prevalence IPP, is defined as

$$IPP := \max_{\tau\in[0,\infty)} I(\tau). \tag{4}$$

A high infected peak prevalence (i.e., a large number of infected people in a given time) may collapse the health systems capacity and the intensive care units (ICU) in hospitals.

Define now $S_\infty := \lim_{\tau\to\infty} S(\tau)$ and $I_\infty := \lim_{\tau\to\infty} I(\tau)$, which for a $\sigma(\cdot) \in \Omega^\sigma$, depend on conditions $S(\tau_f), I(\tau_f)$, and $\overline{\mathcal{R}}$. Then, $I_\infty = 0$ (acute infection) and

$$S_\infty(\overline{\mathcal{R}}, S(\tau_f), I(\tau_f)) := -\frac{W(-\overline{\mathcal{R}}S(\tau_f)e^{-\overline{\mathcal{R}}(S(\tau_f)+I(\tau_f))})}{\overline{\mathcal{R}}}, \tag{5}$$

where $W(\cdot)$ is the Lambert function. So, the second epidemic-severity index, i.e., the epidemic final size EFS, is defined as

$$\begin{aligned} EFS &:= 1 - S_\infty \\ &= 1 + (1/\overline{\mathcal{R}})W(-\overline{\mathcal{R}}S(\tau_0)e^{-\overline{\mathcal{R}}(S(\tau_0)+I(\tau_0))}). \end{aligned} \tag{6}$$

A high epidemic final size means a high total number of infected people at the end of the epidemic. Further details on the general behavior of system (3), including an initial conditions sensitivity analysis, are given in Sereno et al. (2022).

2.3. Control objectives

Here, we introduce a definition of the control objectives based on the two main epidemiological indexes. Lemma 3.1 in Sereno et al. (2022), gives an upper bound for the steady-state value of S for any $\sigma(\cdot)$ belongs to Ω^σ: this supremum is given by S^*, and the conditions for $\sigma(\cdot)$ to achieve it are that $S(\tau_f) \approx S^*$ and $I(\tau_f) \approx 0$, which implies that system (3) achieves a Quasi steady-state (QSS) condition at τ_f. Furthermore, in Sereno et al. (2022) is also stated that is possible to arbitrarily reduce the IPP, while maintaining $S_\infty \approx S^*$, since condition $S_\infty = S^*$ only fixes the integral of I over $[0,\infty)$ - denoted as the area under the curve of $I(\tau)$, $AUC_I := \int_0^\infty I(t)dt$ - but not its shape and peak.

More precisely, we have that $\int_0^\infty I(t)dt = 1 - S_\infty = 1 - S^*$, which implies that no matter which forms $\sigma(\cdot) \in \Omega^\sigma$ takes, to fix S_∞ at S^* fixes only $\int_0^\infty I(t)dt$: arbitrary small values of IPP will be achieved at the price of an arbitrarily long duration of the epidemic ($I(\tau)$ will keep positive for a long time). This is the reason why, we argue, it is not right in an optimal control context to try to minimize $I(\tau)$, or even IPP - as most MPC-based control strategies do (Köhler et al., 2021; Morato et al., 2020; Morris et al., 2021) - since parameter $\overline{\mathcal{R}}$ determines a balance between IPP and the duration of the epidemic, which cannot be modified by NPIs.

Before starting a formal control objective, we need to consider also the side effects of the non-pharmaceutical interventions (i.e., economic, educational, and psychological side effects, among others). They will be assessed by means of the Social Distancing Index, defined as:

$$SDI := \int_0^\infty \sigma(t)dt = \int_{\tau_i}^{\tau_f} \sigma(t)dt. \tag{7}$$

Now, according to the latter discussion, we pose the control objective as follows.

Definition 1 (*Control Objectives Sereno et al. (2022)*). Consider system (3) with initial conditions $(S(0), I(0)) = (1-\epsilon, \epsilon)$, $0 < \epsilon \ll 1$, and $\overline{\mathcal{R}} > \frac{1}{1-\epsilon}$. Consider also that $\sigma(\cdot) \in \Omega^\sigma$, and that a desired maximal value for I, $I_{\max} > 0$, is established according to the health system capacity. Then, the epidemiological control objective consists of bringing the EFS to $1 - S^*$, while maintaining $IPP \le I_{\max}$. In addition, the social/economic control objective consists in minimizing SDI, provided that the Epidemiological Control Objective is achieved.

In Sereno et al. (2022), several theoretical open-loop strategies were proposed ('single interval', 'wait-maintain-suspend', and optimal policies) to account for these two main objectives. The objective of this work is to extend this idea – which is not directly applicable to real-life problems – to realistic feedback schemes that account for uncertainties and more complex descriptions of the epidemic.

3. Switched NMPC

3.1. Sampled system

The first step towards an applicable control strategy is to obtain a discrete-time version of system (3) together with (2), to be able to apply a computer-based strategy as it is model predictive control (MPC). To obtain the general form $x_{k+1} = F(x_k, \sigma_k)$, $k \in \mathbb{I}_{0:\infty}$, where $F(\cdot)$ is the discrete-time non-linear function corresponding to (3), x_k is the state vector at the sampled time k (s.t. $x_k := x(kT_s)$, being $T_s > 0$ the sampling time), and σ_k stands for the input such that $\sigma(\tau) = \sigma_k$, $t \in [kT_s, (k+1)T_s]$, $k \in \mathbb{I}_{0:\infty}$.

3.2. Switched NMPC formulation

In this section, a model-based controller, which takes explicit advantage of the analysis made in previous sections, is designed. The main idea is to extend the optimal control strategy (OC) proposed in Sereno et al. (2022) (which is not a proper feedback control able to reject disturbances) to a realistic MPC scheme.

The cost function to be minimized online by the NMPC controller is then given by:

$$V_N(x;\boldsymbol{\sigma}) = \sum_{j=0}^{N-1} r\|\sigma_j\|^2 + q\|S_j - S^*\|^2, \tag{8}$$

where N is the control horizon, r and q are penalizing positive constants, $x = x_k = (S_k, I_k)$ is the current state at time k and $\boldsymbol{\sigma} := \{\sigma_0, \sigma_1, \dots, \sigma_{N-1}\}$ is the predicted control sequence, with $\sigma_j \in \mathbb{I}_{[0,n]}$, for $j \in \mathbb{I}_{[0,N-1]}$. The optimization problem to be solved at each sampling $k \in [k_i, k_f]$, with $k_i := \frac{\tau_i}{T_s}$, $k_f := \frac{\tau_f}{T_s}$, is then given by:

$$\begin{aligned} \min_{\boldsymbol{\sigma}} \quad & V_N(x;\boldsymbol{\sigma}) \\ s.t. \quad & \\ & x_0 = x_k, \\ & x_{j+1} = F(x_j, \sigma_j), \quad j \in \mathbb{I}_{[0,N-1]}, \\ & x_j \in \mathcal{X}, \quad j \in \mathbb{I}_{[0,N-1]}, \\ & \sigma_j \in \mathbb{I}_{[0:n]}, \quad j \in \mathbb{I}_{[0,N-1]}, \\ & I_j \le I_{\max}, \quad j \in \mathbb{I}_{[0,N-1]}, \end{aligned} \tag{9}$$

where constraint $I_j \le I_{\max}$ is devoted to impeded that the infected population overpasses the maximum $I_{\max}$. Once the optimal solution is computed, the first optimal control action σ_0^* is applied to the system, the discrete-time is shifted forward (i.e., $k \to k+1$), and the iteration continues with a new solution of the optimization problem (receding horizon control (RHC) policy). The use of quantized control actions leads to the use of mixed-integer optimization problem strategies to solve (9).

Remark 1. Formally, there are two forms to achieve the control objectives by means of a receding horizon control (RHC) strategy. The first one (following almost directly the optimal control proposed in Sereno. et al. 2022) is by imposing terminal conditions on S and I such that $S_N = S^*$ and $I_N = I_{det}$, where I_{det} is a small or detectable level, which corresponds to a (quasi) equilibrium condition. This way, the cost function is just devoted to minimizing the interventions, and formal recursive feasibility and stability are guaranteed at the cost of selecting a control horizon N large enough for the terminal constraint

to be feasible.[2] The second one (given by Problem (9)) is by replacing the terminal conditions with the cost term $q\|S_j - S^*\|^2$, with $j = 1, \dots, N$ being the prediction times, and q an appropriated penalization weight. This formulation (theoretically) accepts any control horizon value (since no terminal constraints are included, no recursive feasibility problems appear), but does not formally ensure stability. No conditions on I are needed in this case, since the convergence of S_k to S^* (S is strictly decreasing for any intervention profile) forces I to go to zero, after reaching a maximum, for $k \to \infty$.

Remark 2. Problem (9) is different from that of typical MPC schemes, where the epidemiological control objective is reached by minimizing I over time. Indeed, part of the objectives are achieved by feasibility, not by optimality, since I is not minimized, but constrained to be smaller than $I_{\max}$ along time. This is so because Section 2 shows that I should be left free at the beginning of the epidemic so that the area under its curve reaches the value corresponding to $S_\infty = S^*$. After this transient period, i.e., once $S_\infty = S^*$ and $I_j \le I_{\max}$, the cost function (8) only penalizes the Social Distancing Severity (through σ_j), which is the only thing one can do and represents a non-trivial problem.

3.3. Inclusion of waiting-time constraints

Another critical limitation in epidemic control – usually disregarded by most MPC-based strategies – is the so-called waiting-time constraints (WTC). Non-pharmaceutical intervention (mainly social distancing) cannot be modified every week or maintained for very large periods of time of several months, because of practical/implementation reasons. Indeed, to pretend populations follow this kind of unrealistic social intervention patterns is a major source of uncertainty, producing in some cases dramatic loss of control. To make the control strategy presented in Section 3.2 more realistic, waiting-time constraints, consisting in forcing minimal and maximal periods of time for each social intervention, are explicitly considered. In other words, it is desirable to apply each NPI σ no less than L^σ consecutive times, and no more than U^σ.

Since there are just n possible interventions, there are $n+1$ values for L^σ and $n+1$ values for U^σ (considering the case of no intervention, $\sigma = 0$). However, for $\sigma = 0$, it is assumed $U^0 = \infty$, since the system can be left in an open loop indefinitely. Furthermore, it is reasonable to have more restrictive (smaller) values of U^σ for the harder social intervention (larger values of σ), while the values of L^σ are selected according to social organization limits.

For the sake of simplicity, no further details on the waiting-time constraints are given here. However, the implementation of such constraints in problem (9) is neither direct nor trivial. Indeed, to account for minimal and maximal waiting-time constraints, some new variables accounting for the past implemented strategies are necessary, and some logic constraints need to be included. A detailed analysis of this kind of problem – in the general context of the control of switching systems – is given in Anderson et al. (2021) and Perez, Anderson, Hernandez-Vargas, González, and Actis (2022).

3.4. Properties of the NMPC controller

The main advantages of the proposed sNMPC in contrast to other strategies are summarized next:

(i) As a feedback controller, the control strategy is allowed to react in real-time for the system evolution. This is a critical property to properly help authorities in decision-making, given the uncertainty in the data and the errors in the policy implementation due to people's resistance.

(ii) The EFS is driven to $1 - S^*$, the IPP is maintained under an arbitrarily selected value $I_{\max}$ (epidemiological control objective), while SDI is (as long as possible) minimized (social/economic control objective).

(iii) The way the sNMPC optimization problem is posed allows us to include discrete control actions and WTC, which force the controller to select only realistic policies to be implemented.

(iv) From a theoretical perspective, the proposed sNMPC ensures closed-loop stability, at least when no WTCs are considered (as shown in Rawlings and Risbeck (2017)).

4. Simulations

In this section, the proposed controller is tested through different simulation scenarios. First, the simple SIR model is considered, to compare its performance with the OC presented in Sereno et al. (2022), in an uncertainty scenario. Then, a more detailed model is considered (the SIDARTHE model introduced in Giordano et al. (2020)) to show that the benefits of the proposal are general enough to account for such kinds of extensions. A qualitative comparison is made with other similar strategies and, mainly, waiting-time constraints are included to analyze the versatility of the strategy. Finally, performance analysis for different waiting-time scenarios is carried out using the SIDARTHE model. This later analysis seeks to be useful for authorities aiming to implement reasonable NPIs with minimal and maximal times imposed by social factors.

4.1. Numerical implementation

The optimization problem to be solved at time $k \in \mathbb{I}_{[0,\infty]}$ is given by Eq. (9), which is a mixed-integer problem. For the simulations, however, the reproduction number is expressed as a combination of binary variables that simplified the whole computation and allows us to consider WTC straightforwardly. For each prediction time $j \in \mathbb{I}_{[0,N-1]}$, the reproduction number reads

$$\begin{aligned} \mathcal{R}^{\sigma_j} &= \overline{\mathcal{R}} - \sigma_j \Delta\mathcal{R} \\ &= \overline{\mathcal{R}} - \mathbf{U} \cdot \mathbf{v}_j, \end{aligned}$$

where,

$$\mathbf{U} = [0,\ \Delta\mathcal{R},\ 2\,\Delta\mathcal{R},\ \dots\ ,n\,\Delta\mathcal{R}],$$
$$\mathbf{v}_j = [v_j^0,\ v_j^1,\ v_j^2, \dots, v_j^n]^T,$$

with

$$v_j^\ell = \begin{cases} 1, & \textit{if NPI } \ell \textit{ is active at } j \\ 0, & \textit{otherwise} \end{cases}, \quad \text{for } \ell \in \mathbb{I}_{[0,n]},$$

fulfilling the following condition:

$$\sum_{\ell=0}^{n} v_j^\ell = 1, \quad \text{for} \quad j \in \mathbb{I}_{[0,N-1]}. \tag{10}$$

Note that $\mathbf{v}_j$ (a vector with binary components) is devoted to selecting which NPI is active at each prediction time and, by condition (10), only one of them can be active at a time.

For the WTC we introduce two new setting parameters to the optimization problem. First, the vector parameter $\mathbf{v}_{-1}$, which fulfills condition (10) as the optimization variable $\mathbf{v}_j$, $j \in \mathbb{I}_{[0,N-1]}$, allows us to particularize which NPI severity was active at time $k-1$. Second, the vector parameter $\mathbf{H}_{-1}$ allows us to count how many consecutive periods of time the NPI indicated by $\mathbf{v}_{-1}$ has been active. Then, the following Big-M constraint is added to the optimization problem (9) to be solved at time k,

$$\mathbf{v}_{-1}^T \cdot (\mathbf{WT}_{\min} - \mathbf{H}_{-1}) \le \mathbf{B}_m \cdot (\mathbf{v}_{-1}^T \cdot \mathbf{v}_0), \tag{11}$$

where $\mathbf{WT}_{\min} = [L^0, L^1, \dots, L^n]^T$ is a vector containing the $n+1$ values of L^σ, and $\mathbf{B}_m$ is the Big-M constant (a large constant forcing the constraint

[2] N should respect the time I needs to reach its maximum and then decrease to zero.

fulfilling). Note that $\mathbf{v}_0$, i.e., the first optimization variable along the prediction horizon, is the only decision variable in constraint (11). In order to consider the minimal waiting-time constraints all along the prediction horizon, $\mathbf{v}_j$ with $j \in \mathbb{I}_{1:N-1}$, the following dynamical variable is included in the optimization problem (9):

$$\mathbf{H}_{j+1} = (\mathbf{H}_j + \mathbf{v}_j) \odot \mathbf{v}_j, \qquad j \in \mathbb{I}_{[0,N-1]}, \tag{12}$$

where $\mathbf{H}_0 = \mathbf{H}_{-1}$, and the operator $\odot$ is the element-wise product, also now as the Hadamard product Marcus and Khan (1959). Finally, the following Big-M constraint is included in the optimization problem (9),

$$\mathbf{v}_j^T \cdot (\mathbf{WT}_{\min} - \mathbf{H}_{j+1}) \leq \mathbf{B}_m \cdot (\mathbf{v}_j^T \cdot \mathbf{v}_{j+1}), \; j \in \mathbb{I}_{[0,N-1]}. \tag{13}$$

To account for the maximal waiting-times we define $\mathbf{WT}_{\max} = [U^0, U^1, \dots, U^n]^T$ as a vector that contains each of the $n+1$ values of U^σ, and include the next constraint in the optimization problem,

$$\mathbf{H}_{j+1} \leq \mathbf{WT}_{\max}, \quad j \in \mathbb{I}_{[0,N-1]}, \tag{14}$$

where $\mathbf{H}_j$ is defined by Eq. (12).

The simulations were run on a laptop computer with an i7-4510U processor (2 cores, 4 threads, 2.0–3.1 GHz) and 16 GB RAM under MATLAB R2021a and CasADi version 3.5.5 Andersson, Gillis, Horn, Rawlings, and Diehl (2019), using the BONMIN solver.

4.2. Basic SIR model

In this section, to emphasize the benefits of the proposed closed-loop strategy in contrast to the open-loop optimal control (OC) strategy proposed in Sereno et al. (2022) - the basic System (3) is considered, under uncertainty scenarios. The uncertainty is included by means of multiplicative noise in the states (so that the larger the compartment populations, the bigger the noise in the state measurement) and additive noise in the control action.

In the first case, where the noise describes the typical uncertainty of government epidemiological data, the estimated states are assumed to be given by $\hat{I} = I \cdot \eta_I$, where η_I is a normally distributed noise with mean 1 and standard deviation 0.3, (i.e., $\eta_I \sim \mathcal{N}(1, 0.3)$). Furthermore, $\hat{S} = S + I \cdot (1 - \eta_I)$, which allows us to maintain the assumption of constant population.

In the second case, where the noise describes the possibility that the population does not comply with the government interventions, the control actions to be applied to the system are assumed to be given by

$$\hat{\mathcal{R}}^\sigma(\tau) = \begin{cases} \mathcal{R}^\sigma(\tau), & \text{for } \sigma(\tau) = 0 \\ \mathcal{R}^\sigma(\tau) + \eta^\sigma, & \text{for } \sigma(\tau) \in \mathbb{I}_{[1:n]}, \end{cases} \tag{15}$$

where η^σ is again a normally distributed noise with mean 0.3 and standard deviation 0.3, i.e. $\eta^\sigma \sim \mathcal{N}(0.3, 0.3)$. The selected noise distributions systematically underestimate the effectiveness of a given NPI to describe, as said, the disregarding of government interventions. Furthermore, no disturbances are considered for $\sigma(\tau) = 0$, since no interventions have not uncertainty.

We simulate a time period of 365 days, with a sampling time $T_s = 5$ days, considering an IPP of $I_{\max} = 0.07$. $\overline{\mathcal{R}}$ is assumed to be 3.0 (i.e., $\beta = 0.30$ days^{-1} and $\gamma = 0.1$ days^{-1}), while $\underline{\mathcal{R}} = 0.5$, which means that $\Delta\mathcal{R} = \frac{\overline{\mathcal{R}} - \underline{\mathcal{R}}}{n} = 0.625$ for $n = 4$ ($\mathcal{R}^\sigma$ may assume the following five values: $\mathcal{R}_0 = 3.0$, $\mathcal{R}_1 = 2.375$, $\mathcal{R}_2 = 1.75$, $\mathcal{R}_3 = 1.125$ and $\mathcal{R}_4 = 0.5$). According to $\overline{\mathcal{R}}$ value, the herd immunity takes a value of $S^* = 1/3$. The initial state is given by $I(0) = 1.0 \times 10^{-3}$, $S(0) = 1 - I(0)$, while the sNMPC controller works from $\tau_i = 15$ to $\tau_f = 180$ days, and then the system is left in open loop for remaining of the year. The control horizon of the MPC is selected to be $N = 40$ days, while the penalties r and q are selected to be 1×10^{-1} and 1×10^3 (to prioritize the EFS against the SDI index), respectively. In this first scenario, no WTC is considered for the sNMPC, to make a fair comparison with the OC (that has not included this kind of constraint). Although not shown for the sake of brevity, it is worth mentioning that the performance of both controllers under nominal conditions (no disturbances, no model-plant mismatches) are, as expected, quite similar. However, significant differences appear when more realistic scenarios are simulated.

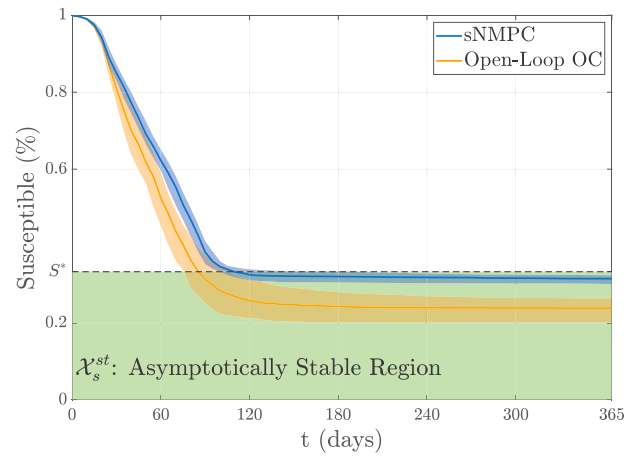

Fig. 3. Fraction of susceptible population for SIR model. Blue and yellow color represent the sNMPC and the Open-Loop OC, respectively. Solid lines and shaded areas represent the mean values and the lower–upper quartiles (Q1–Q3), respectively.

Table 2
Performance indexes for the SIR model.

Index	Open-loop OC	sNMPC
EFS	0.7612	0.6869
IPP	0.0963	0.0761
SDI	30.6136	50.5219

Fig. 3 shows the susceptible population throughout a year under the open-loop OC and the proposed sNMPC, for 100 random uncertainty realizations. Although both controllers steer the value of the susceptible population within the stable region $\mathcal{X}_s^{st}$, only the sNMPC is able to reach the optimal value S^*, due to feedback reaction (indeed, the OC steers the susceptible to a significantly low value). This is highly desired since the IPP (the maximal fraction of infected individuals over time) will be reduced as long as S_∞ is maintained in the vicinity of S^*.

The evolution of the infected population is shown in Fig. 4. Note that with the proposed controller, the infected population is close to the ICU capacity, so a potential collapse of the health system is avoided. Instead, with the OC, although infectious cases decrease faster, there is a peak that significantly exceeds the ICU capacity, which may increase the number of deaths.

Finally, Fig. 5 shows the time evolution of $\mathcal{R}^\sigma$ for both controllers. As can be seen, the OC is not as aggressive as the sNMPC, for it does not account for the significant model mismatch by feedback. As such, even when I is far over $I_{\max}$ and S is far under S^*, the OC does not react as desired.

For the sake of numerical comparison, the performance index corresponding to each controller is shown in Table 2.

4.3. SIDARTHE model

One of the main advantages of the dynamical characterization of SIR-type systems made in Sections 2 and 3 is that it is general enough to describe many extended versions of the basic model. One of the most widely used is the so-called SIDARTHE mathematical model (Giordano et al., 2020). This dynamical system (developed for Italy) consists of eight ordinary differential equations, describing the evolution of the population in each stage over time. A particularization of this model –

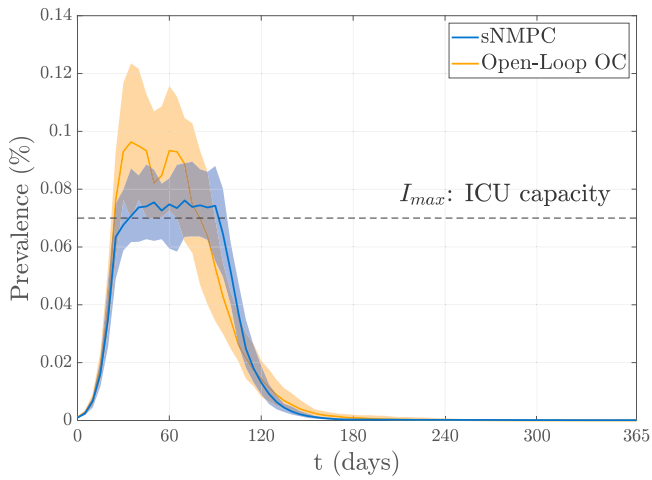

Fig. 4. Fraction of infected population for SIR model. Blue and yellow color represent the sNMPC and the Open-Loop OC, respectively. Solid lines and shaded areas represent the mean values and the lower–upper quartiles (Q1–Q3), respectively.

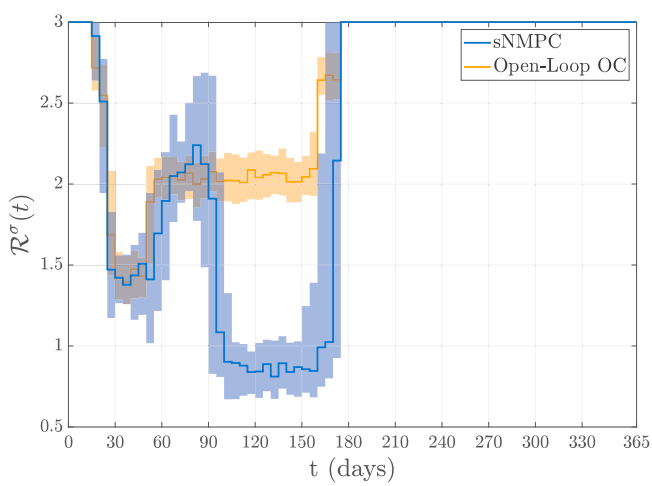

Fig. 5. Reproduction number as a control action for SIR model simulation scenario. Blue and yellow color represent the sNMPC and the Open-Loop OC, respectively. Solid lines and shaded areas represent the mean values and the lower–upper quartiles (Q1–Q3), respectively.

developed for Germany and used to test an MPC controller Köhler et al. (2021) – is as follows:

$$\begin{aligned}
\dot{S} &= -S(\alpha I + \beta D + \gamma A + \beta R),\\
\dot{I} &= S(\alpha I + \beta D + \gamma A + \beta R) - (\epsilon + \zeta + \lambda)I,\\
\dot{D} &= \epsilon I - (\zeta + \lambda)D,\\
\dot{A} &= \zeta I - (\theta + \mu + \kappa)A,\\
\dot{R} &= \zeta D + \theta A - (\mu + \kappa)R,\\
\dot{T} &= \mu A + \mu R - (\sigma + \tau)T,\\
\dot{H} &= \lambda I + \lambda D + \kappa A + \kappa R + \sigma T,\\
\dot{E} &= \tau T,
\end{aligned} \tag{16}$$

where Susceptible (S), Infected (I), Diagnosed (D), Ailing (A), Recognized (R), Threatened (T), Healed (H), and Extinct (E) are the state variables and represent the fraction of population in each stage. The meaning and values of the system parameters are shown in Table 3. The detection rate of asymptomatic cases is considered negligible, that is, the policymakers only test people that develop symptoms, i.e., $\epsilon = 0$ see Köhler et al. (2021). The key point to analyzing this kind of extended model is by grouping its different compartments into three

Table 3
Parameters of the SIDARTHE model Köhler et al. (2021).

Parameter	Value (days^{-1})	Meaning
$\alpha_{\min}$	0.0422	Transmission rate between Susceptible and Infected
$\alpha_{\max}$	0.3614	
$\gamma_{\min}$	0.0422	Transmission rate between Susceptible and Ailing
$\gamma_{\max}$	0.3614	
β	0.0084	Transmission rate between Susceptible and Diagnosed
θ	0.1981	Probability rate of detection for symptomatic cases
ζ	0.0790	Probability rate at which an unaware infected subject develops clinically relevant symptoms.
μ	0.0130	Rate at which infected individuals, in A or R, develop life-threatening symptoms.
τ	0.0401	Mortality rate.
λ	0.0596	Rate of recovery for infected
κ	0.0563	Rate of recovery for ailing
σ	0.0922	Rate of recovery for threatened

main categories: Susceptible, Infected/Infectious, and Removed. This way, we have that *Susceptible* is given by S, while *Infectious* is given by $\{I, D, A, R, T\}$, and *Removed* is given by $\{H, E\}$. Then, as long as only one flow is considered from S to one of the subdivisions of the infected (in the SIDARTHE case, S is connected only with I) we can make the same stability analysis of the previous section. Indeed, this reasoning is also valid for any other epidemic model fulfilling the aforementioned condition, independently of the Removed subdivisions and their corresponding flows coming from the Infected ones. Additionally, since the *Infected* compartment is divided into 5 sub-compartments (to have a better description of real cases), the maximum constraint imposed on I in the Optimization Problem (9) can be now imposed only to the threatened compartment T. This relaxation significantly improves the controller performance, as shown in the next simulations.

The reproduction number of the SIDARTHE model is given by,

$$\mathcal{R}(\cdot) = \frac{1}{\zeta + \lambda}\left(\alpha(\cdot) + \frac{1}{\theta + \mu + \kappa}\left(\gamma(\cdot)\zeta + \frac{\beta\theta\zeta}{\mu + \kappa}\right)\right),$$

where now, the time-varying parameters are α and γ, which are the parameters affected by the NPI measures. This way, the minimum and maximum values, $\underline{\mathcal{R}}$ and $\overline{\mathcal{R}}$, are computed according to the minimum and maximum values of these parameters (see Table 3).

Similarly to the SIR case, we consider state multiplicative noise in the Infected, Ailing, and Recognized population, of 50%, 20%, and 1%, respectively. More precisely, $\hat{I} = I \cdot \eta_I$, with $\eta_I \sim \mathcal{N}(1, 0.5)$, $\hat{A} = A \cdot \eta_A$, with $\eta_A \sim \mathcal{N}(1, 0.2)$, and $\hat{R} = R \cdot \eta_R$, with $\eta_R \sim \mathcal{N}(1, 0.01)$. Also, to maintain the total population constant, $\hat{S} = S + I \cdot (1 - \eta_I) + A \cdot (1 - \eta_A) + R \cdot (1 - \eta_R)$. The SIDARTHE reproduction number is used as the control action to be applied to the system assuming the same noise structure reported by Eq. (15). In the SIDARTHE case η^{σ} is a normally distributed noise with mean 0 and standard deviation 0.05, i.e. $\eta^{\sigma} \sim \mathcal{N}(0, 0.05)$ (note that this uncertainty may result in an under or overestimation of the control action).

As before, we simulate a time period of 365 days, with a sampling time $T_s = 5$ days. Threatened individuals T, are constrained to be smaller or equal than $T_{\max} = 0.01$. Furthermore, according Table 3, we have $\overline{\mathcal{R}} = 3.429$, while $\underline{\mathcal{R}} = 0.4456$. Therefore, for $n = 4$ we have that $\Delta\mathcal{R} = 0.7459$ and $\mathcal{R}^{\sigma}$ may assume the following five values: $\mathcal{R}_0 = 3.429$, $\mathcal{R}_1 = 2.6832$, $\mathcal{R}_2 = 1.9373$, $\mathcal{R}_3 = 1.1915$ and $\mathcal{R}_4 = 0.4456$. The initial state is given by $I(0) = 6.0241 \times 10^{-6}$, $A(0) = 3.6627 \times 10^{-6}$, $R(0) = 5.7831 \times 10^{-7}$, and $S(0) = 1 - I(0) - A(0) - R(0)$, while the sNMPC controller works from $\tau_i = 15$ to $\tau_f = 180$ days, and then the system is left in open loop for remaining of the year. The control horizon of the MPC is selected to be $N = 30$ days, while the penalties r and q are selected to be 1×10^{-1} and 1×10^{3} (to prioritize the EFS against the SDI index), respectively. The waiting time bounds, L^{σ} and U^{σ}, are selected as $L^{\sigma} = 15$ days, for $\sigma \neq 0$, and $U^{\sigma} = 30$ days, for $\sigma \neq 0$. For $\sigma = 0$ (no intervention case) the bounds are selected to be $L^0 = T_s$ and $U^0 = \infty$.

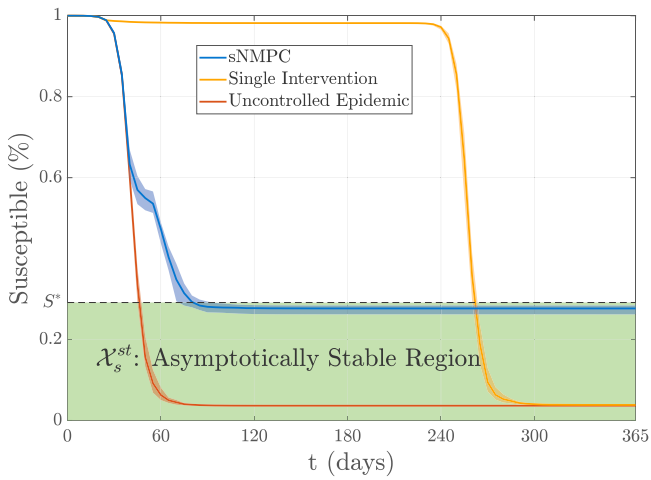

Fig. 6. Fraction of susceptible population for SIDARTHE model. Blue, yellow, and red colors represent the sNMPC, Single Intervention, and Uncontrolled Outbreak, respectively. Solid lines represent mean values, shaded areas represent lower–upper quartiles (Q1–Q3).

Table 4
Performance indexes for the SIDARTHE model.

Index	Uncontrolled	Single-intervention	sNMPC
EFS	0.9632	0.9613	0.7219
IPP	0.4899	0.4629	0.2868
SDI	0	106.8120	79.8655

To assess the sNMPC performance, a (rather typical) single-interval strategy (consisting of a six months full lockdown strategy, from $\tau_i = 30$ to $\tau_f = 210$ days, $\mathcal{R}^\sigma(\tau) = \underline{\mathcal{R}}$) and the uncontrolled epidemic case ($\mathcal{R}^\sigma(\tau) = \overline{\mathcal{R}}$, for all τ) are also simulated and plotted, for 100 randomly uncertainty realizations. Figs. 6–9 show the temporal evolution of Susceptible, Total Infected, Threatened, and Extinct individuals, respectively, while Fig. 10 shows the NPI severity, i.e., $\mathcal{R}^\sigma(t)$. We observe that the sNMPC performs well on average for all the uncertainties considered. That is, the susceptible population remains in the vicinity of the optimal value $S^* = 0.2916$, while both, the single-interval intervention and the uncontrolled epidemic, go to a significantly smaller value ($S_\infty \approx 0.01$). Similar behavior to that of these later cases is observed under the MPC strategy proposed in Köhler et al. (2021). In addition, the sNMPC controller is able to maintain the Threatened population under the IPP, as shown in Fig. 8. On the other hand, Fig. 9 shows that the sNMPC achieves a lower value for the Extincted population, in comparison with the single-interval, and the uncontrolled epidemic scenarios. Finally, Fig. 10 shows that the sNMPC achieves the epidemiological objective with a significantly smaller socio-economic cost than the one corresponding to the (hard) single interval strategy while fulfilling the WTC. For the sake of numerical comparison, the performance index corresponding to each strategy is shown in Table 4.

Remark 3. The main claim in Sereno et al. (2022) (and Sections 2.1 and 2.3) is that directly minimizing the infected compartment (without accounting for the EFS) is not a good practice in the context of an optimal-type controller (as it is the MPC) since this leads to a not well-posed problem. Given that Köhler et al. (2021) (as many other studies proposing MPC strategies, as Carli et al., 2020; Morato et al., 2020; Péni et al., 2020 and Péni & Szederkényi, 2021) follows this line, a direct comparison with our controller is not possible. Indeed, for all the mentioned controllers, the EFS ($1 - S_\infty$) is systematically (and significantly) greater than $1 - S^*$.

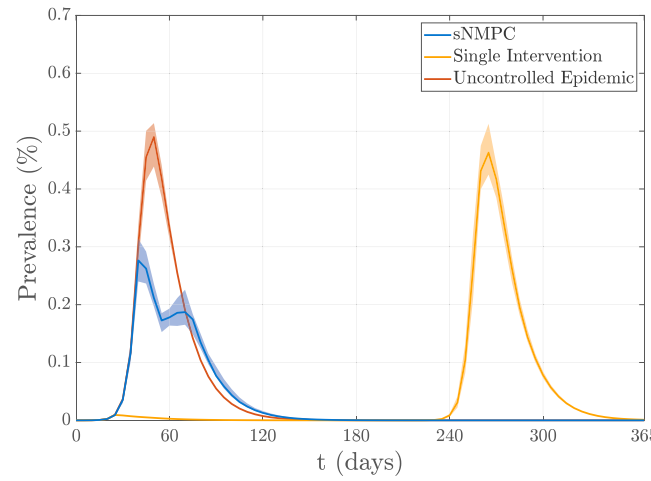

Fig. 7. Fraction of total infected population for SIDARTHE model ($I + D + A + R + T$). Blue, yellow, and red colors represent the sNMPC, Single Intervention, and Uncontrolled Outbreak, respectively. Solid lines represent mean values, shaded areas represent lower–upper quartiles (Q1–Q3).

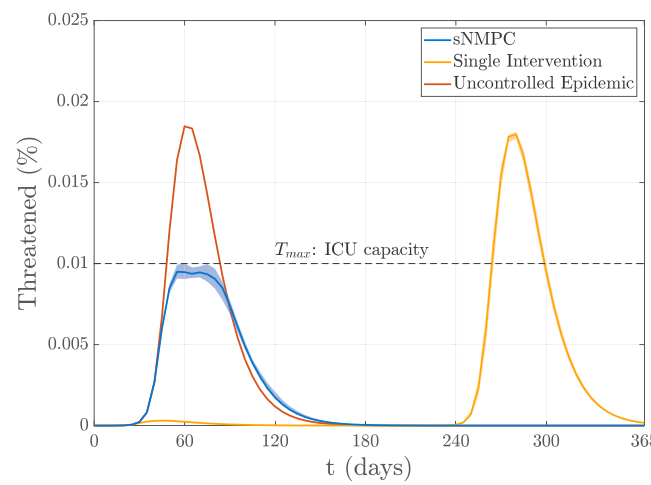

Fig. 8. Fraction of Threatened population for SIDARTHE model. Blue, yellow, and red colors represent the sNMPC, Single Intervention, and Uncontrolled Outbreak, respectively. Solid lines represent mean values, shaded areas represent lower–upper quartiles (Q1–Q3).

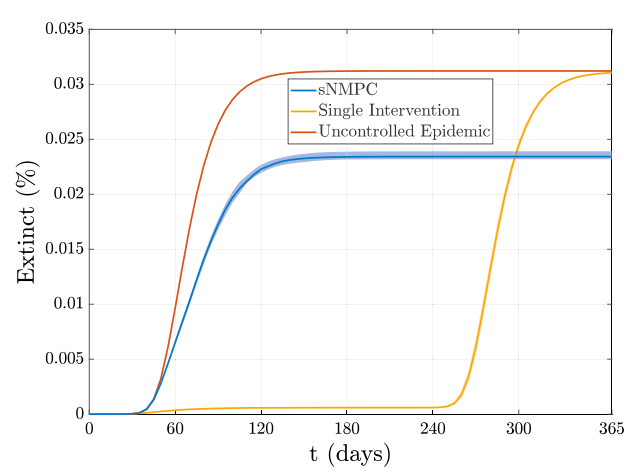

Fig. 9. Fraction of extinct population for SIDARTHE model. Blue, yellow, and red colors represent the sNMPC, Single Intervention, and Uncontrolled Outbreak, respectively. Solid lines represent mean values, shaded areas represent lower–upper quartiles (Q1–Q3).

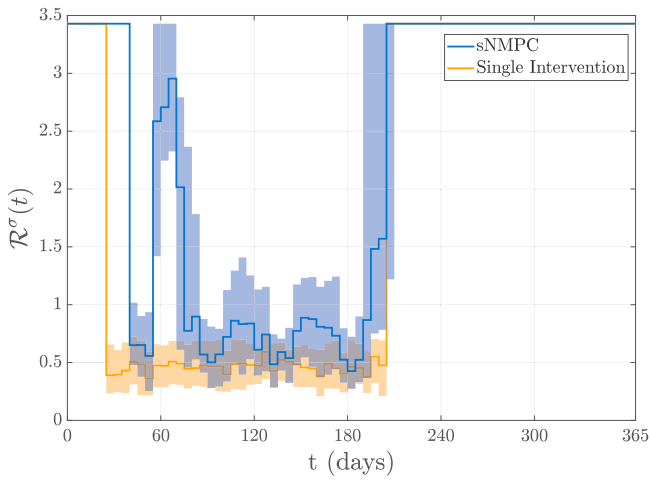

Fig. 10. Reproduction number as a control action for SIDARTHE model. Blue, and yellow colors represent the sNMPC and the Single Intervention scenario, respectively. Solid lines represent mean values, shaded areas represent lower–upper quartiles (Q1–Q3).

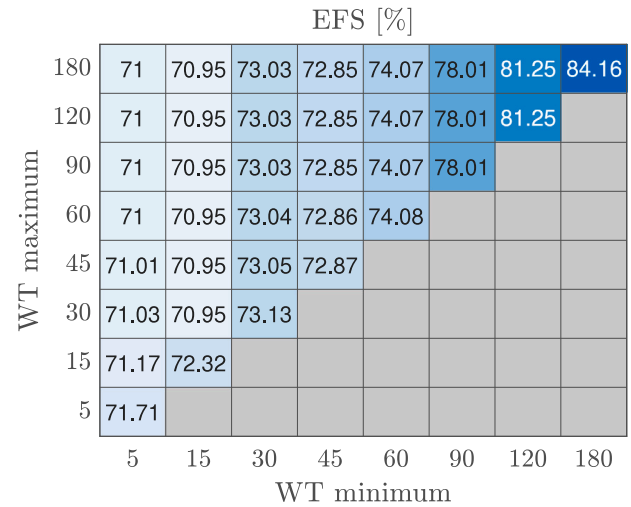

Fig. 11. Heat maps reporting the Epidemic Final Size in the percentage of the population. The index is defined as the smaller the better. Infeasible combinations of min–max dwell-time constraints are represented by gray box values.

4.4. Performance analysis

To analyze the performance of the proposed controller in terms of the waiting-time constraints, the EFS and SDI indices defined in Eqs. (6) and (7), respectively, will be studied through several simulations of the SIDARTHE model (the IPP is not considered, since it shows no significant variations as it accounted for a constraint). To do that, we considered all the combinations of L^σ and U^σ (fulfilling the condition $L^\sigma \leq U^\sigma$) in the following range of values: $\{5, 15, 30, 45, 60, 90, 120, 180\}$ days. According to Definition 1, minimizing the EFS is the primary objective, while minimizing SDI is the secondary one, so the SDI index will be analyzed after the EFS.

The heat map in Fig. 11 shows the EFS (epidemiological index) for each feasible pair of WTC, the darker blue meaning the higher (worst) values. In general, better epidemiological performances (lower values of EFS) will be obtained for large values of U^σ and small values of L^σ, since this way WTCs vanish. Reciprocally, given that $L^\sigma \leq U^\sigma$ by definition, worse performances will be obtained for L^σ approaching U^σ from below. After a first glance at Fig. 11, we can see that the EFS is more sensible to the minimum than to maximum WTC variations. In general, larger minimum WTCs correspond to larger EFS, although the increase of this latter is not linear, given the non-linearity of the closed-loop system (note that there is not a significant difference between the EFS corresponding to minimum WTC of 15 to 5 days, suggesting that not extremely short time periods T_s are needed).

It is noteworthy that the value of the EFS does not increase significantly for a maximum WTC greater than 60 days. This result has an important connotation since many governments chose to apply quarantines that were too long, and this is not directly related to an increase in performance, i.e., to a reduction of the total infected population. The best performance scenario is achieved with $L^\sigma = 15$ days and $U^\sigma = 30$ days; the worst, with $L^\sigma = 180$ days and $U^\sigma = 180$ days. This later case corresponds to the case in which all NPI measures change (exactly) every six months, which results in a second-wave outbreak. To assess the difference between the best and worst scenarios, if we consider a population of 80 *million* of people, in one of the scenarios there will be approximately 10 *million* more people infected than in the other. This highlights the importance of properly selecting this type of WTC in the optimal control problem formulation.

The heat map in Fig. 12 shows the SDI for each feasible pair of WTC, the darker blue meaning the higher (worst) values. The best value is achieved with $L^\sigma = 15$ days, and similarly to EFS, the SDI values increase as the minimum WTC increases. However, the increase is not monotone, given the nonlinearity of the closed-loop system.

In general, we observe that for minimum WTC values greater than or equal to 30 days, as the maximum WTC increases, the SDI index increases. This behavior suggests that the fact that $\mathbf{WT}_{\min} \geq 30$ days forces the sNMPC to apply stronger measures to counteract the lack of flexibility to quickly change between different NPI measures. When for minimum WTC is selected to be 5 and 15 days, we observe that as the maximum WTC increases the SDI decreases; moreover, for $\mathbf{WT}_{\max} > 30$ the SDI index keeps almost fixed in a given value. This behavior seems to suggest that lower values of minimum WTC allow the controller to quickly change the NPI measures, and therefore quickly reach the epidemiological control objective without needing a maximum WTC greater than 30 days. For this index, the best performance scenario is reached with $L^\sigma = 15$ days and $U^\sigma = 30$ days, while the worst is reached with $L^\sigma = 120$ days and $U^\sigma = 120$ days and leads to a second wave.

An interesting case of the behavior of SIR-type systems under WTC is the appearance of second-wave patterns. As we can see in Figs. 13(a) and 13(b), as the minimum WTC increases, the second-wave pattern tends to become more pronounced. In this case, as the minimum WTC increases the sNMPC has less capability to control the epidemic (indeed, the system is less controllable), and opts to apply longer and harder NPI measures that generate these second-wave patterns. In spite of the second wave appearance, the controller guarantees the accomplishment of the epidemiological control objectives, i.e., S reaches S^* and the threatened population is kept under $T_{\max}$. Figs. 13(a) and 13(b), show the threatened population behavior and the applied NPI measure, $\mathcal{R}^\sigma(t)$, for minimal WTC values of 15, 30, 60, and 120 days.

5. Conclusion

In this work, based on previous dynamical analysis of controlled SIR-type models, a Switched Model Predictive Control is proposed, to help authorities to schedule non-pharmaceutical policies in epidemic contexts. Opposite to most optimizing strategies in the literature, we consider different control objectives from the minimization of the infected and/or the severity of the control actions. This way, the proposed strategy achieves – even under uncertain/disturbed scenarios – the minimum of EFS, while maintaining the IPP under an upper bound, generally specified by the healthcare capacity. The extra degree of freedom of the optimizing strategy is used to minimize the side effect of interventions. Another distinctive property of the proposal – out of the

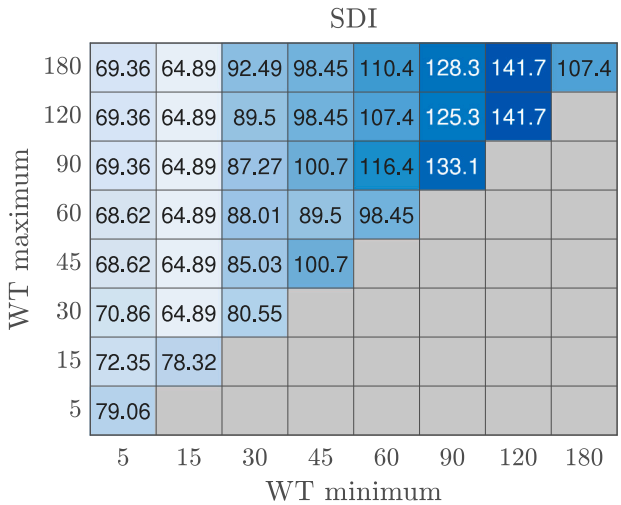

Fig. 12. Heat maps reporting the Social Distancing Index. The index is defined as the smaller the better. Infeasible combinations of min–max dwell-time constraints are represented by gray box values.

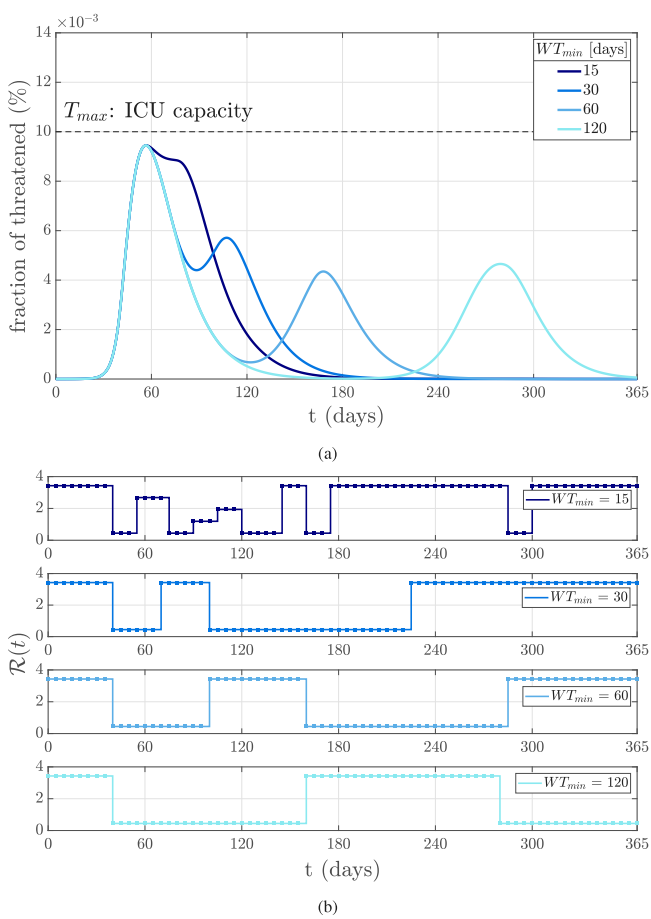

Fig. 13. (a) Threatened population behavior vs changes in WT minimum. A longer WT minimum generates a second-wave behavior in the epidemiological evolution. (b) Control action pattern vs changes in WT minimum. Here we consider four different levels of WT minimum, 15, 30, 60, and 120 days.

use of discrete levels of interventions – is the inclusion of maximal and minimal time limits for the application of each particular NPI, which allows policymakers to better understand non-intuitive behaviors of epidemics under intervention time constraints.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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